

Industrial Automation

“COLLABORATIVE ROBOTS Help Create Jobs, Not Steal Them”



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At EFY Conferences in March this year, there was a talk about integration of a new generation of robot workers in the labour workforce in order to increase productivity and efficiency. Known as ‘Cobots,’ short for Collaborative Robots, these robots are a combination of industrial robotics and automation. Pradeep David, general manager, India & Sri Lanka, Universal Robots, speaks to Dilin Anand from EFY about the role of cobots in production floors and the impact on human workforce

Q. Unmanned manufacturing—does it really make sense now that robots have arrived?

A. The idea is for humans and robots to be inter-dependent and achieve what each of them does best, safely. There are a few things requiring human ingenuity that are best done manually, whereas accuracy, precision and repetitive mass production at higher efficiency are best taken care of by robots. Cobots provide an added incentive to the workforce as consistent quality production is possible with human supervision.

Q. What sets cobots apart?

A. Collaborative robots are lightweight, cost-effective, safe and easy to operate. The human-robot collaboration is different from that between traditional robots. With cobots, the days of hiring an expensive external consulting team every time a robot has to be programmed are over. The new reality is that even operators with no programming experience can quickly program the cobot arms (using intuitive, 3D visualisation).

Q. How easy are they to operate?

A. All the user has to do is move the robot arm to the desired waypoints or touch arrow keys on an

easy-to-use touchscreen tablet. The cobot arms come with an average payback time of 195 days. That’s the fastest in the industry—quite simply because they are void of all the added costs traditionally associated with automation, such as external programming resources and shielded work cells. So automation can no longer be considered out of reach now.

Q. Which applications stand to benefit most?

A. Production setups today often need to be flexible and agile in order to meet changing market demands and stay competitive. The lightweight cobot arms can be easily moved and re-deployed to new processes, enabling users to automate virtually any manual task, including those with small batches or fast changeovers. They can work right next to humans without any safety guard, thus making robotics technology accessible to all levels of the industry. Cobots are ideally suited for jobs that human operators find repetitive and dull.

Q. What differentiates them from conventional industrial and robotic arms?

A. Industrial robots are caged to keep humans protected. Service

robots are meant to safely leave the cage while doing tasks for humans. Collaborative robots, on the other hand, are ‘force limited’ robots—their built-in technology allows them to work safely alongside humans (subject to safety assessments of the application). So these work more as a ‘worker’s assistant’ in the form of a ‘portable tool,’ unlike traditional industrial robots. Collaborative robots come in all sizes and shapes, and have integrated sensors and soft and rounded surface to ensure safety of human workforce and reduce the risk of damage due to impact, pinching and crushing.

Q. There are fears that robots will take away human jobs. With the rise of cobots, what is the future like?

A. Instead of replacing human workforce and causing job losses, collaborative robots help companies expand operations, thereby creating jobs and many more opportunities at supervision level. A human-machine study conducted by MIT researchers at a BMW factory has found that teams comprising humans and robots collaborating efficiently can be around 85 per cent more productive than teams having either humans or robots alone. **EFY**